**CM25D008**

# PRACTICAL NO :-07

**AIM** :- A system has RAM which can store four pages simultaneously to satisfy
page requests. Write a program to calculate percentage page hit and page fault if
the system uses Least recently used page replacement technique for the page
references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

MINGW64:/c/Users/sejal sarode/OneDrive/Desktop

```
  GNU nano 8.7
#include <stdio.h>

int main() {
    int frames = 3;
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int n = 12;

    int frame[3], time[3];
    int i, j, k, pos, min;
    int pageFaults = 0, pageHits = 0;

    for(i = 0; i < frames; i++) {
        frame[i] = -1;
        time[i] = 0;
    }

    for(i = 0; i < n; i++) {
        int found = 0;

        for(j = 0; j < frames; j++) {
            if(frame[j] == pages[i]) {
                pageHits++;
                time[j] = i;
                found = 1;
                break;
            }
        }

      // If not found → PAGE FAULT
        if(!found) {
            min = time[0];
            pos = 0;

            for(j = 1; j < frames; j++) {
                if(time[j] < min) {
                    min = time[j];
                    pos = j;
                }
            }

            frame[pos] = pages[i];
            time[pos] = i;
```

```c
        // If not found → PAGE FAULT
        if(!found) {
            min = time[0];
            pos = 0;

            for(j = 1; j < frames; j++) {
                if(time[j] < min) {
                    min = time[j];
                    pos = j;
                }
            }

            frame[pos] = pages[i];
            time[pos] = i;
            pageFaults++;
        }
    }

    printf("Total Page Hits = %d\n", pageHits);
    printf("Total Page Faults = %d\n", pageFaults);

    float hitRatio = (float)pageHits / n * 100;
    float faultRatio = (float)pageFaults / n * 100;

    printf("Hit Percentage = %.2f%%\n", hitRatio);
    printf("Fault Percentage = %.2f%%\n", faultRatio);

    return 0;
}
```

```
sejal sarode@INBOOK_X2_SLIM MINGW64 ~/OneDrive/Desktop
$ gcc lru.c -o lru

sejal sarode@INBOOK_X2_SLIM MINGW64 ~/OneDrive/Desktop
$ ./lru
Total Page Hits = 2
Total Page Faults = 10
Hit Percentage = 16.67%
Fault Percentage = 83.33%
```